Mini-project 1 in Advanced Machine Learning (02460)

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Part A: VAE Priors

We train VAEs with a Bernoulli likelihood on binarised MNIST, using three priors: a standard Gaussian, a 10-component MoG, and a flow (8 affine coupling layers). Encoder and decoder are symmetric 3-layer MLPs (784–512–512, $M=32$, ReLU); for non-Gaussian priors the KL is estimated by Monte Carlo.

Figure 1 shows expected differences: the Gaussian prior sits as single probability mass over the posterior, where the MoG and Flow priors reshape themselves to track the digit clusters. The test ELBOs however barely separate (Gaussian -88.8 ± 0.7, MoG -88.2 ± 0.7, Flow -88.1 ± 0.4; mean±std over 5 runs), indicating the shared decoder is the bottleneck: when reconstruction saturates, a more expressive prior has little to improve. The flow prior trains the most stable (lowest variance).

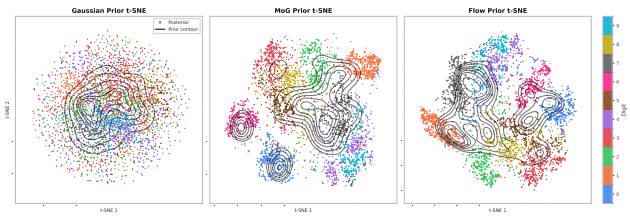

Figure 1: t-SNE projections of aggregate posterior (coloured by digit class) and prior density contours for Gaussian (left), MoG (middle), and Flow (right) priors.

Part B: Sampling Quality

We compare three generative pipelines on continuous MNIST. The **DDPM** pairs a five-level U-Net (32, 64, 128, 256, 256 channels, skip connections, LogSigmoid, $T=100$) with a linear noise schedule. The **latent DDPM** first trains a β-VAE (same MLP backbone as Part A but with a Gaussian likelihood, $M=32$), then fits a fully-connected DDPM ($T=1000$) to the encoder means. At test time it draws latent samples from the diffusion model and pushes them through the frozen decoder.

Pixel-space DDPM gives the best FID (36.96) but only manages 35 samples/s (Table 1). The Gaussian VAE is 370× faster and not far behind in FID (42.30). Latent DDPM runs diffusion in 32 dimensions instead of 784, increasing throughput to ∼8k samples/s, though the best FID (81.91 at $\beta=10^{-3}$) lags both baselines. The choice of β matters: too small (10^{-6}) and the posterior is diffuse, giving the latent DDPM a hard target (FID 115), too large (≥ 0.1) and reconstruction quality drops (FID >88).

Figure 2: 2×2 samples: VAE Gaussian (left), DDPM U-Net (mid), Latent DDPM $\beta=10^{-3}$ (right).

Table 1: FID (↓) and sampling speed on standard MNIST ($N=1000$).

Model	FID	Samp./s
VAE (Gaussian)	42.30	13,179
DDPM (U-Net)	36.96	35
Lat. DDPM $\beta=10^{-6}$	115.15	8,405
Lat. DDPM $\beta=10^{-3}$	81.91	8,613
Lat. DDPM $\beta=0.1$	88.80	8,584
Lat. DDPM $\beta=1$	153.18	8,861

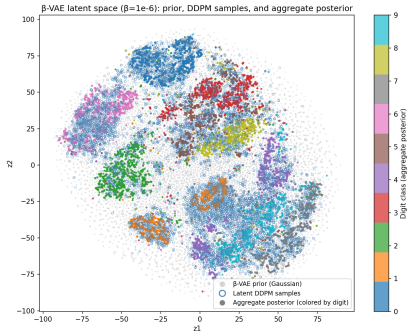

Figure 3: Latent space ($\beta=10^{-6}$): Gaussian prior (grey), DDPM latents (blue), aggregate posterior (coloured).

Fig. 3 shows the latent space at $\beta=10^{-6}$: the aggregate posterior (coloured) splits into ten digit clusters that the Gaussian prior (grey) does not match. The DDPM distribution (blue) covers the posterior mass much better, but overall FID remains high (115 vs. 42 for the VAE) because at this β the decoder itself is the limiting factor.

Code snippets

Declaration of use of generative AI

This declaration **must** be filled out and included as the 3rd page of the document. The questions apply to all parts of the work, including research, project writing, and coding.

- I/we have used generative AI tools: [yes / no]

If you answered *yes*, please complete the following sections. List the generative AI tools you have used:

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Describe how the tools were used:

What did you use the tool(s) for?

At what stage(s) of the process did you use the tool(s)?

How did you use or incorporate the generated output?

References